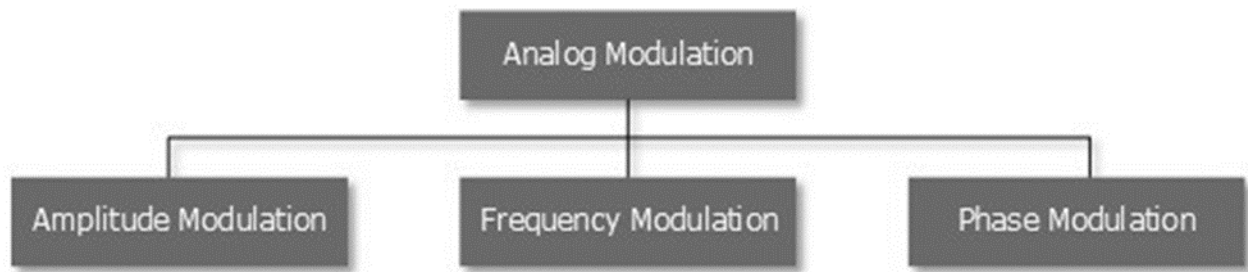


ANALOG MODULATION SYSTEMS



Amplitude Modulation

In this modulation, the amplitude of the carrier signal is modified to reflect the analog data. Let the carrier frequency be given by

$$x_c(t) = A_c \cos \omega_c t$$

and the modulating signal be denoted by $x_m(t)$. Then the modulated signal $X_{AM}(t)$ is given by

$$X_{AM}(t) = (A_c + x_m(t)) \cos \omega_c t = A_c \cos \omega_c t + x_m(t) \cos \omega_c t$$

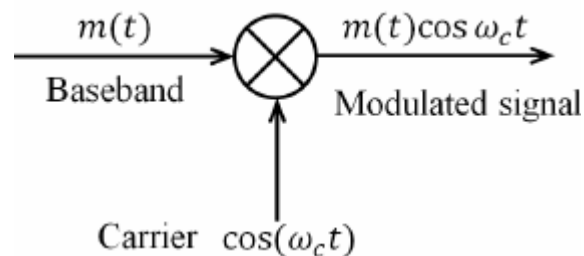
Define modulation index $\mu = \frac{A_m}{A_c}$, then using a fixed frequency modulating signal such that

$$x_m(t) = A_m \cos \omega_m t$$

Then

$$x_{AM}(t) = (A_c + A_m \cos \omega_m t) \cos \omega_c t$$

$$x_{AM}(t) = A_c (1 + \mu \cos \omega_m t) \cos \omega_c t$$



When $\cos\omega_m t$ is maximum, it will be +1 and when it is minimum it will be -1. Consequently by denoting the maximum value and minimum value of $x_{AM}(t)$ by V_{max} and V_{min} respectively, we can write

$$V_{max} = (A_c + \mu)\cos\omega_c t$$

$$V_{min} = (A_c - \mu)\cos\omega_c t$$

Dividing these two quantities at the peak of the carrier, i.e when $\cos\omega_c t = 1$ we obtain the expression

$$\mu = \frac{A_{max} + A_{min}}{A_{max} - A_{min}}$$

Thus it is possible to determine modulation index from the frequency measurements. Further more from the figure:

$$A_c = \frac{A_{max} + A_{min}}{2}$$

And

$$A_m = \frac{A_{max} - A_{min}}{2}$$

Frequency Components

The process of AM changes the amplitude of the carrier. The carrier itself is a single, pure tone – but what frequency components are present?

Substituting a single-tone modulation $m(t) = A_m \cos\omega_m t$ into the AM generation equation, we have

$$x_{AM}(t) = A_m \cos\omega_m t \cos\omega_c t + A_c \cos\omega_c t$$

$$x_{AM}(t) = A_c \cos\omega_c t + A_m \cos\omega_m t \cos\omega_c t$$

The second term is the product of two sinusoids. Using the trigonometrical expansion for $\cos\alpha\cos\beta$ followed by the substitutions $\alpha \rightarrow \omega_c t$ and $\beta \rightarrow \omega_m t$, we end up

$$A_m \cos\omega_m t \cos\omega_c t = \frac{1}{2} [\cos(\omega_c t + \omega_m t) + \cos(\omega_c t - \omega_m t)]$$

The AM wave is therefore of the form of

$$x_{AM}(t) = A_c \cos\omega_c t + \frac{A_m}{2} [\cos(\omega_c t + \omega_m t) + \cos(\omega_c t - \omega_m t)]$$

Evidently, the spectrum for a fixed amplitude modulating signal $m(t)$ at frequency ω_m results in a frequency component at ω_c with amplitude A_c as well as at $\omega_c \pm \omega_m$ with amplitude $\frac{A_m}{2}$.

The former is the result of adding the carrier, and the latter is, indirectly, the result of the multiplication of carrier and modulation.

Since $\mu = \frac{A_m}{A_c}$, the amplitude $\frac{A_m}{2}$ may be written as $\frac{\mu A_c}{2}$ – that is, it is proportional to the modulation index μ .

Power Analysis

As well as frequency and related bandwidth considerations, the issue of power is important.

More power implies the need for larger capacity subsystems such as the output electronics and antenna. But more power consumption implies higher cost and shorter battery life for portable transmitters. Even if we do use more power, we want to be sure that the power is

actually performing a useful function. The previous discussion on the spectrum of AM indicates that a substantial amount of power is used just in transmitting the carrier, and this does not actually help transmit the modulation signal itself.

To analyze the power and efficiency of AM, first recall that for a peak amplitude A the RMS value of a waveform is $V_{rms} = A/\sqrt{2}$. We found earlier that the carrier amplitude in AM is A_c and the sidebands each have an amplitude of $A_m/2 = \mu A_c/2$. The total AM signal power is thus

$$\begin{aligned} P_{total} &= P_{carrier} + 2 \times P_{sidebands} \\ &= A_c^2 + 2 \times \left(\frac{\mu A_c}{2} \right)^2 \\ &= P_{carrier} \left(1 + \frac{\mu^2}{2} \right) \end{aligned}$$

It is reasonable to define the efficiency as the power in the sidebands (which actually transmit “information”) divided by the total power consumed and is thus

$$\begin{aligned} \eta &= \frac{\text{Power in the sidebands}}{\text{Total power}} \\ &= \frac{\mu^2}{2 + \mu^2} \end{aligned}$$

It follows that when $\mu = 0$, the efficiency is zero. In that case, the power is completely in the carrier (and there is no modulation, which is not useful at all). However, when $\mu = 1$, the efficiency is 1/3. This shows that the efficiency of AM is not especially good, and as noted earlier, a large proportion of the power is used in simply transmitting

the carrier. Thus, we reach the conclusion that AM is not a very efficient scheme in terms of power efficiency, as much of the power is wasted in transmitting the carrier.

Question

The sidebands in AM carry significant power, and their power depends on the modulation index.

a) Write an equation for the power in the carrier in terms of the carrier amplitude A_c .

b) Write an equation for the power in each sideband in terms of the sideband amplitude A_m .

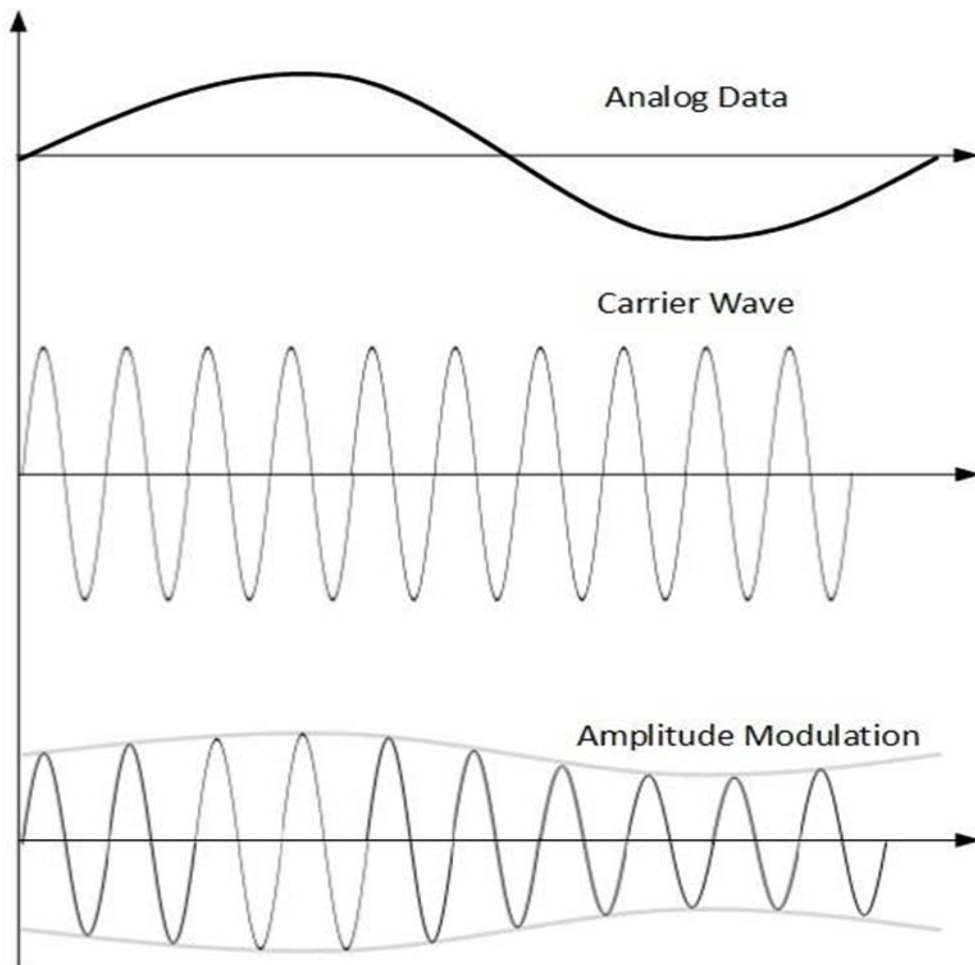
c) Using the above results, show that the total power present in an AM waveform is the carrier power multiplied by a factor of

$$\left(1 + \mu^2/2\right)$$

d) Defining the efficiency η as the power in the sidebands divided by the total power, show that

$$\eta = \frac{\mu^2}{2 + \mu^2}$$

Comment on the efficiency when the modulation index is zero, and when the modulation index is unity.



Amplitude modulation is implemented by means of a multiplier. The amplitude of modulating signal (analog data) is multiplied by the amplitude of carrier frequency, which then reflects analog data. The frequency and phase of carrier signal remain unchanged.

Angle Modulation

In angle modulation, the angle of the carrier $\phi(t)$ is varied linearly with the message signal

$$\phi(t) = 2\pi f_c t$$

$$f = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

Or in general we can write

$$f = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

Or frequency is the rate of change of the angle.

Angle modulation may be classified into two:

Phase modulation: In phase modulation, the angle of the carrier is varied linearly with the message

$$\phi_i(t) = 2\pi f t + k_p m(t)$$

The phase modulated wave is therefore given by

$$S_{PM}(t) = A \cos(2\pi f t + k_p m(t))$$

Frequency modulation: In which the frequency of the carrier is varied linearly with the message signal $m(t)$

$$f = f_c + k_f m(t)$$

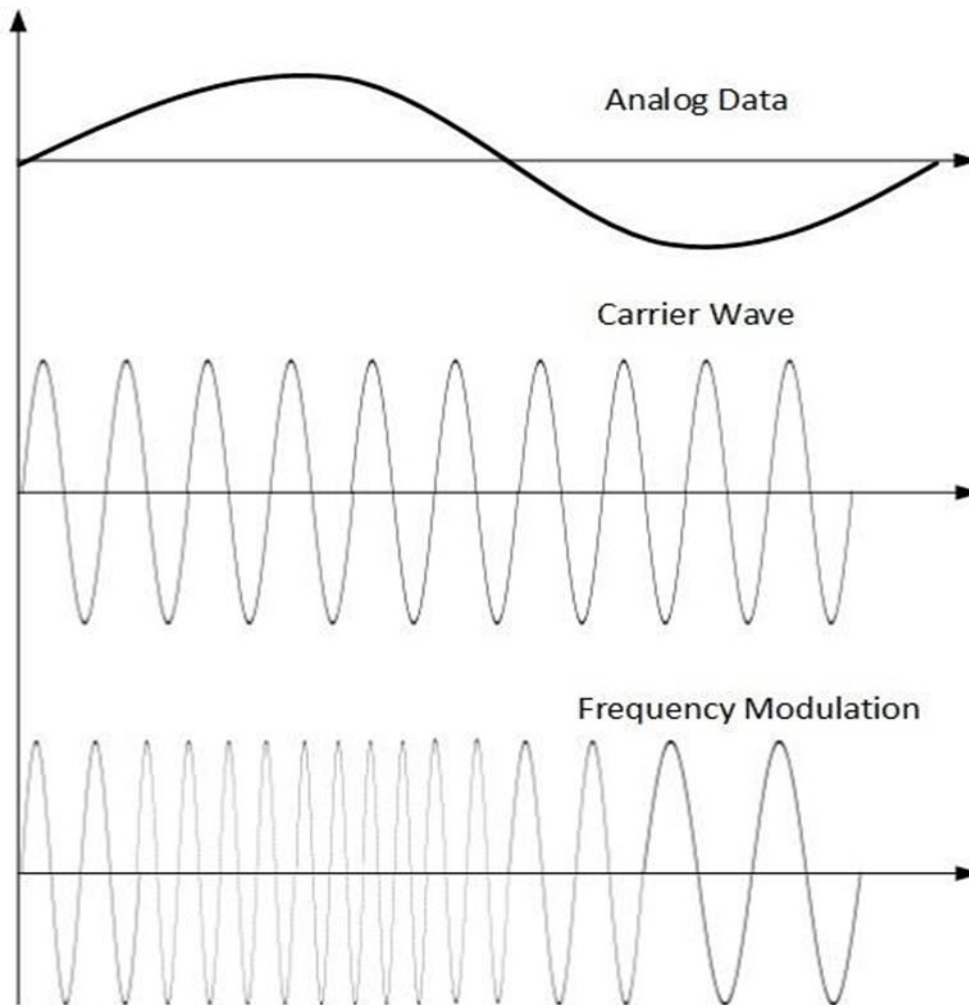
$$Q(t) = \int 2\pi f dt = 2\pi f_c t + 2\pi k_f \int m(t)$$

The phase modulated waveform is therefore given by

$$S_{PM}(t) = A \cos(2\pi f_c t + 2\pi k_f \int m(t))$$

Frequency Modulation

In this modulation technique, the frequency of the carrier signal is modified to reflect the change in the voltage levels of the modulating signal (analog data). The amplitude and phase of the carrier signal are not altered.



The amplitude and phase of the carrier signal are not altered.

Phase Modulation

In the modulation technique, the phase of carrier signal is modulated in order to reflect the change in voltage (amplitude) of analog data signal.

